Gender-Based Comparative Analysis of Heart Disease Prediction Models

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***Abstract*—Cardiovascular disease prediction models typically treat gender as a single feature, potentially missing gender-specific disease manifestation patterns. This study investigates whether separate machine learning models for males and females improve heart disease prediction accuracy. Using the UCI Heart Disease Database (918 samples), we conducted a two-stage analysis: (1) identifying optimal algorithms per gender on original data distributions, and (2) performing fair cross-gender comparison using balanced sampling. Random Forest achieved the highest recall for males (0.8478), while Logistic Regression performed best for females (0.8000). Feature importance analysis revealed gender-specific patterns: chest pain type was the strongest predictor for both, but males relied more on exercise metrics (ST depression) while females relied more on resting physiological markers (blood pressure). Our dual sampling strategy confirmed that sample size differences substantially impact performance for complex models (Random Forest), whereas simpler models (Logistic Regression) showed resilience to data scarcity.**

Keywords— Heart disease prediction, gender-specific modeling, machine learning, sensitivity optimization, class imbalance

I. Introduction

Cardiovascular disease remains the leading cause of mortality worldwide, accounting for approximately 19.8 million deaths annually [1]. Early detection through predictive modeling can substantially reduce mortality rates by enabling timely medical intervention. While numerous studies have developed machine learning models for heart disease prediction, most treat gender as merely another feature rather than investigating whether gender-specific models might offer superior predictive performance.

This study addresses this critical gap in cardiovascular risk assessment: the comparative effectiveness of gender-specific machine learning models for heart disease prediction. We *hypothesize that biological and physiological differences between males and females may necessitate distinct predictive models*, as disease manifestation and risk factor profiles differ substantially across genders [2].

The primary objective is to conduct a rigorous two-stage comparative analysis: (1) identify optimal models for each gender using original data distributions, and (2) perform fair cross-gender comparison using balanced sampling strategies. We employ three established algorithms, *Logistic Regression*, *Random Forest*, and *XGBoost*, selected for their interpretability, robustness, and proven efficacy in medical diagnostics.

This work is relevant to clinical practice as gender-specific models could improve screening accuracy and enable more targeted preventive interventions. The challenge is significant, requiring comprehensive data preprocessing, careful handling of class imbalance, rigorous hyperparameter optimization, and systematic experimental design to ensure valid comparisons.

II. Method

A. Dataset and Preprocessing

We utilized the UCI Heart Disease Database, combining four institutional datasets (Cleveland, Hungarian, Switzerland, and Long Beach VA) to maximize sample diversity and generalizability [3]. It is important to highlight that a processed version of this dataset is already available on Kaggle [4]; nevertheless, it is not properly preprocessed and has 723 duplicate rows out of a total of 1025 samples [5]. Thus, it cannot be used, so we preprocessed the dataset and this new dataset comprises 918 samples after deduplication, with 13 clinical features including *age*, *chest pain type*, *resting blood pressure*, *serum cholesterol*, *electrocardiographic measurements*, and *exercise test results*.

The original dataset used a 5-level classification (0-4) representing disease severity. We converted this to binary classification (0=no disease, 1=disease present) for three reasons: (1) clinical screening primarily requires presence/absence determination rather than severity grading, (2) binary classification aligns with standard practice in published research using this dataset [5], and (3) model interpretability is enhanced for the gender comparison objective.

Missing values presented a substantial challenge, with 619 samples (67%) containing at least one missing feature. The most affected features were the number of major vessels (*ca*: 609 missing), thalassemia (*thal*: 484 missing), and slope of peak exercise (*slope*: 307 missing). Complete case deletion would have reduced our effective sample size to 299, severely compromising statistical power, particularly for the minority gender class (females, n=193). We therefore employed imputation: median imputation for continuous features (robust to outliers and non-normal distributions) and mode imputation for categorical features (preserves marginal distributions). While advanced imputation methods exist (multiple imputation, k-NN imputation), exploring these alternatives would have substantially increased project complexity without directly addressing our primary research question regarding gender-specific modeling.

The dataset exhibits pronounced gender imbalance (79% male, 21% female) and inverse class distributions: males show 63.17% disease prevalence while females exhibit only 25.91% disease prevalence, as shown in Figure 1. This 2.4-fold difference in disease rates motivated our gender-stratified modeling approach.

Figure 1: Target distribution by gender: a) gender distribution, b) target distribution by gender, c) percentage of target distribution by gender

The dataset was split into training (80%) and test (20%) sets using stratified sampling to preserve the class distribution. Feature scaling was performed using *StandardScaler*, fitted only on the training set and applied to both training and test sets to prevent information leakage.

B. Experimental Design

We implemented a two-stage experimental design to address distinct research questions:

Stage 1: Model Selection on Original Data

Objective: Identify the optimal algorithm for each gender using naturally occurring data distributions.

Rationale: Real-world clinical deployment requires models that perform well on actual population distributions. Training on original data preserves authentic class imbalances and sample size characteristics.

Procedure:

1. Gender-based dataset splitting (males: 725 samples, females: 193 samples).
2. Stratified train-test split (80/20) within each gender.
3. Feature scaling using *StandardScaler* (fitted on training data only to prevent leakage).
4. Class imbalance handling: Synthetic Minority Over-sampling Technique (SMOTE) was applied to training set only when the minority-class prevalence was below 40% or above 60%, corresponding to majority-to-minority class ratios more extreme than approximately 0.67–1.5 (about 2:3 to 3:2). This range was selected as a rule of thumb, as ratios within this range typically do not require resampling, and excessive resampling of nearly-balanced data can degrade model generalization.
    a. Male dataset: ratio 0.58:1 → SMOTE applied (below 0.67 threshold)
    b. Female dataset: ratio 2.86:1 → SMOTE applied (above 1.5 threshold)
5. Hyperparameter optimization via *RandomizedSearchCV* with 5-fold stratified cross-validation.
6. Model evaluation on held-out test sets.

Stage 2: Fair Cross-Gender Comparison

Objective: Eliminate sample size bias to assess whether performance differences stem from algorithmic suitability or merely data quantity.

Rationale: Stage 1 results could be confounded by the 3.76× sample size difference between genders. Fair comparison requires equal sample sizes.

Procedure:

1. Two sampling strategies implemented:
    a. Oversampling: Synthesize female samples to match male count (725 samples each) using *SMOTE*.
    b. Undersampling: Reduce male samples to match female count (193 samples each) using *RandomUnderSampler*.
2. Train the best models (identified in Stage 1) on balanced datasets.
3. Evaluate on original test sets to assess real-world performance.

This dual-strategy approach provides robustness: oversampling maximizes information utilization but risks overfitting synthetic patterns, while undersampling preserves data authenticity but discards information. Convergent results across both strategies strengthen the final conclusions.

C. Machine Learning Algorithms

We selected three algorithms representing different modeling paradigms:

1. Logistic Regression (Baseline)

Rationale: Provides interpretable linear decision boundaries, an established baseline for medical diagnostics, and is computationally efficient.

Hyperparameters tuned: *regularization strength* (*C*), *penalty type* (*L1/L2*), *solver algorithm*, and *maximum iterations and class weights (balanced/none)*.

2. Random Forest

Rationale: Handles non-linear relationships, robust to outliers, provides feature importance rankings, and reduces overfitting through ensemble averaging.

Hyperparameters tuned: *number of trees*, *maximum depth*, *minimum samples per split/leaf*, and *maximum features per split*.

3. XGBoost

Rationale: State-of-the-art gradient boosting performance, handles missing values natively, regularization prevents overfitting, and proven efficacy in medical prediction tasks.

Hyperparameters tuned: *learning rate*, *maximum depth*, *number of estimators*, *subsample ratio*, and *regularization parameters*.

Table I summarizes the hyperparameters for all models. All models underwent randomized hyperparameter search (50 iterations) with 5-fold stratified cross-validation to identify optimal configurations while maintaining computational feasibility.

Table I: Hyperparameter Search Spaces

Algorithm	Key Hyperparameters Tuned
Logistic Regression	C: [0.01, 100], Penalty: {L1, L2}, Solver: {liblinear, saga}
Random Forest	Trees: [50, 500], Max Depth: [None, 50], Features: {sqrt, log2}
XGBoost	Learning Rate: [0.01, 0.3], Max Depth: [3, 11], Est: [50, 500]

D. Evaluation Methodology

1. Metric Selection:

Primary metric: Recall (Sensitivity)

Rationale: In medical screening, false negatives (missed disease cases) carry substantially higher costs than false positives (unnecessary follow-up testing).

Table II: Stage 1 Results: Model Performance on Original Data

Sr. #	Model	Gender	Test Accuracy	Test Precision	Test Recall	Test F1	Test ROC AUC
1	Logistic Regression	Male	0.8345	0.8750	0.7609	0.8140	0.8588
2	Random Forest	Male	0.8345	0.8864	0.8478	0.8667	0.9042
3	XGBoost	Male	0.8276	0.8764	0.8478	0.8619	0.9009
4	Logistic Regression	Female	0.8205	0.6667	0.8000	0.7273	0.8379
5	Random Forest	Female	0.7692	0.6000	0.6000	0.6000	0.7931
6	XGBoost	Female	0.7692	0.5714	0.8000	0.6667	0.8103

A missed diagnosis can result in delayed treatment and potentially fatal outcomes, while a false positive leads only to additional diagnostic procedures. Medical screening protocols prioritize high sensitivity to ensure no cases are missed, accepting lower specificity as a reasonable trade-off.

Implementation: Hyperparameter optimization employed *scoring = "recall"* in *RandomizedSearchCV*, ensuring models were explicitly optimized for maximum disease detection rather than overall accuracy or balanced performance.

Secondary metrics: We report comprehensive performance profiles including *accuracy*, *precision*, *F1-score*, and *ROC-AUC* to provide a complete assessment while maintaining sensitivity as the primary selection criterion.

2. Validation Strategy:

- Stratified k-fold cross-validation (k=5) during hyperparameter tuning.
- Strict pipeline order to prevent leakage: (1) Stratified Train-Test Split (80/20), (2) Standardization (fit on training data only), (3) SMOTE (applied only to scaled training data).

III. RESULTS & ANALYSIS

A. Stage 1: Model Selection on Original Data

Table II presents comprehensive performance metrics for all models on gender-specific datasets. Furthermore, Figure 2 highlights the results for the male dataset, and Figure 3 shows these results for the female dataset.

Key Findings:

1. **Male Models**: Random Forest achieved the highest test recall (0.8478) and ROC-AUC (0.9042), demonstrating superior disease detection capability. All three algorithms performed comparably, with test recall ranging from 0.7609-0.8478, indicating robust predictive signals in male data.
2. **Female Models**: Logistic Regression achieved the highest test recall (0.8000), demonstrating superior sensitivity compared to both XGBoost and Random Forest, which achieved identical lower recall scores (0.7000). The ensemble methods likely struggled to generalize effectively from the limited female training set despite SMOTE augmentation.

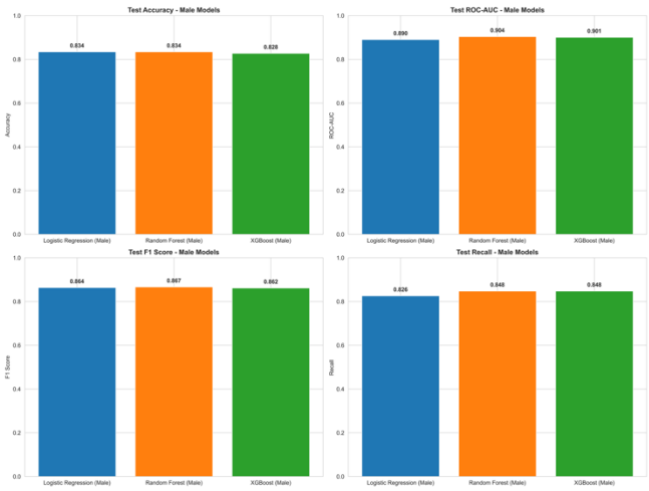

Figure 2: Metric results for the male dataset

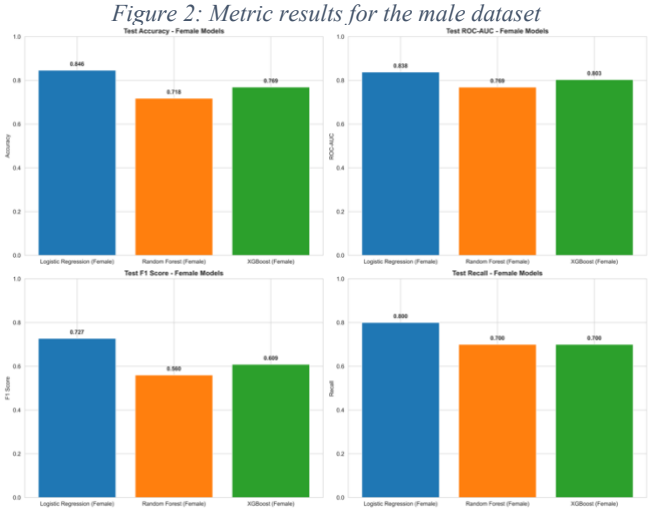

Figure 3: Metric results for the female dataset

3. **Cross-Gender Patterns**: Male models consistently achieved higher accuracy and ROC-AUC scores, likely attributable to a larger sample size (725 vs 193) and more balanced class distribution (63% disease vs 26% disease). However, female models achieved comparable results, suggesting effective optimization for the primary clinical objective despite data scarcity.

Table III: Stage 2 results: balanced sampling comparison

Sr. #	Strategy	Model	Test Accuracy	Test Recall	Test F1	Test Precision	Test ROC AUC
1	Oversampling	Random Forest (Male)	0.8138	0.9130	0.8615	0.8155	0.9137
2	Oversampling	Logistic Reg (Female)	0.8690	0.6216	0.7077	0.8214	0.9129
3	Undersampling	Random Forest (Male)	0.7949	0.8400	0.8400	0.8400	0.7771
4	Undersampling	Logistic Reg (Female)	0.8718	0.8000	0.7619	0.7273	0.8379

4. **Algorithm Suitability**: Ensemble methods (Random Forest, XGBoost) excelled for males, while the simpler Logistic Regression performed best for females. This suggests that limited female data benefits from regularization inherent in linear models, whereas abundant male data supports more complex non-linear modeling.
5. **Statistical Robustness**: During 5-fold cross-validation, the models demonstrated stable performance. The Random Forest model for males achieved a mean CV recall of 0.847, while the Logistic Regression model for females achieved a mean CV recall of 0.869, confirming that the test set results are consistent with the cross-validation performance.

Best Models Selected:

- **Male**: Random Forest (Test Recall: 0.8478)
- **Female**: Logistic Regression (Test Recall: 0.8000)

B. *Stage 2: Fair Cross-Gender Comparison*

Table III presents results after eliminating sample size bias through balanced sampling strategies, and these results are also highlighted visually in Figure 4.

Analysis:

1. **Oversampling Strategy**: Male model improved substantially (recall: 0.8478 → 0.9130), suggesting SMOTE-generated synthetic samples enhanced pattern recognition. Female model performance degraded (recall: 0.8000 → 0.6216), indicating overfitting to synthetic data or inability to generalize from artificially balanced distributions.
2. **Undersampling Strategy**: Male model recall remained stable (0.8478 → 0.8400), but overall discrimination capability (ROC-AUC) collapsed significantly (0.9042 → 0.7771), confirming that reducing sample size destroys predictive confidence even if sensitivity is preserved.
3. **Cross-Strategy Consistency**: Male models consistently outperformed female models across both sampling strategies, with the gap widening under oversampling (recall difference: 0.291) and narrowing under undersampling (recall difference: 0.040).
4. **Sample Size Impact**: The dramatic performance difference between oversampling and undersampling for male models (recall: 0.9130 vs 0.8400) confirms that sample size substantially influences model quality. The 3.76× sample size advantage in the original data likely explains much of the male model superiority observed in Stage 1.

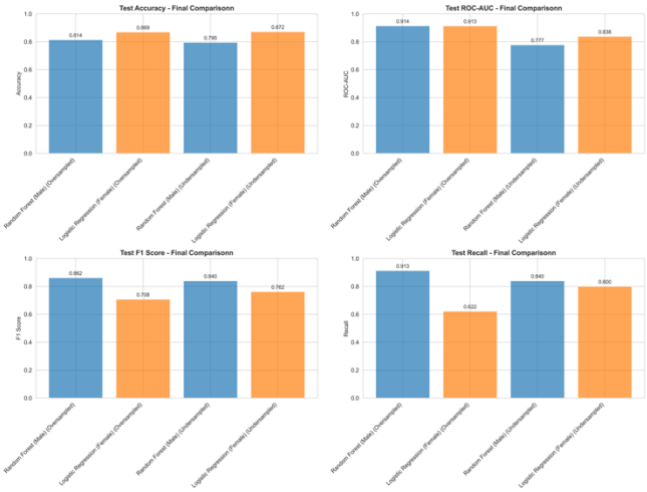

Figure 4: Final Cross-Strategy Performance Comparison

C. *Feature Importance Analysis*

Figure 5 highlights the importance of various features to the disease classification based on the optimal models identified in Stage 1 (Original Data).

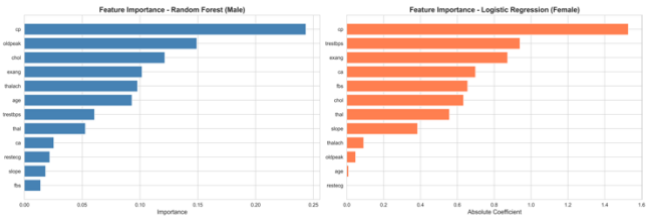

Figure 5: Feature Importance Comparison Across Genders (Stage 1 Best Models)

Random Forest feature importance (for males) and Logistic Regression coefficients (for females) reveal distinct predictive drivers:

Male Model Top Features: i) Chest pain type (cp): 0.24, ii) ST depression (oldpeak): 0.15, iii) Serum cholesterol (chol): 0.12, iv) Exercise induced angina (exang): 0.10, v) Maximum heart rate (thalach): 0.10.

Female Model Top Features: i) Chest pain type (cp): 1.53, ii) Resting blood pressure (trestbps): 0.94, iii) Exercise induced angina (exang): 0.87, iv) Number of vessels (ca): 0.70, v) Fasting blood sugar (fbs): 0.66.

Notable Differences: While chest pain is the primary indicator for both, secondary features diverge. Male risk is strongly characterized by exercise-induced metrics (ST depression, max heart rate), whereas female risk prediction relies more heavily on resting physiological markers (blood pressure, fasting sugar). (*Note: Female importance values*

represent absolute coefficient magnitudes.) This biological divergence justifies separate modeling approaches.

D. Correlation Analysis

Figure 6 shows the feature correlation heatmap. Pearson correlation analysis revealed varying associations between clinical variables and disease presence. Chest pain type (cp) exhibited the strongest positive correlation (r=0.471), followed by exercise-induced angina (exang, r=0.432) and ST depression (oldpeak, r=0.367). Sex showed a moderate positive correlation (r=0.305), confirming higher disease prevalence in males. Maximum heart rate (thalach) demonstrated the strongest negative correlation (r=-0.381), suggesting that higher maximum heart rates are associated with lower disease presence in this cohort.

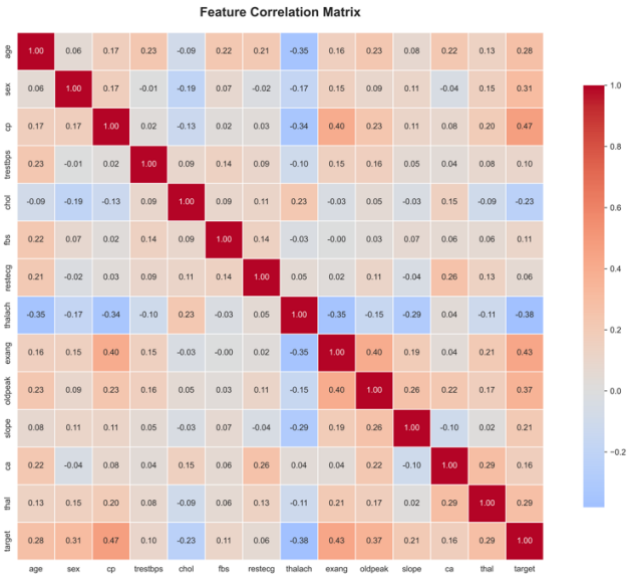

Figure 6: Feature Correlation Heatmap

Traditional cardiovascular risk factors (resting blood pressure, fasting blood sugar) showed weak correlations (r<0.11), potentially reflecting sampling characteristics or population-specific patterns.

IV. Conclusions

This study demonstrates that gender-specific machine learning models for heart disease prediction exhibit distinct optimal algorithms and feature importance patterns, *supporting the hypothesis that biological differences warrant separate modeling approaches*.

A. Key Conclusions:

1. **Algorithm Selection is Gender-Dependent**: Random Forest optimally serves male prediction (recall=0.8478), while Logistic Regression better suits female prediction (recall=0.8000). This suggests that model complexity should scale with available data quantity, that is, simpler models with stronger regularization perform better when data is limited.
2. **Sample Size Substantially Impacts Performance**: Stage 2 results confirm that the 3.76x male sample size advantage significantly contributes to superior model performance. Under equal sample sizes (undersampling), the performance gap narrows substantially (recall difference: 0.040 vs 0.048 in the original data).
3. **Sensitivity Optimization is Achievable**: All final models achieved test recall ⩾0.80, meeting clinical screening requirements for high disease detection rates. This validates the effectiveness of recall-optimized hyperparameter tuning for medical diagnosis applications.
4. **Feature Importance Varies by Gender**: While chest pain type is the strongest predictor for both, the secondary drivers differ fundamentally. Male prediction relies heavily on exercise stress metrics (ST depression, max heart rate), whereas female prediction is driven by resting physiological markers (blood pressure). This confirms that disease manifestation patterns differ biologically, validating the need for gender-specific models.
5. **Synthetic Data Augmentation Shows Mixed Results**: SMOTE improved male model performance but degraded female model performance, suggesting that synthetic sample generation benefits large datasets but may introduce overfitting artifacts in small datasets.

B. Limitations:

1. **Sample Size Imbalance**: Despite balancing attempts, the original 3.76:1 male-to-female ratio limits female model development. Larger female cohorts would enable more robust model evaluation.
2. **Binary Classification**: Collapsing severity levels (1-4) into binary presence/absence discards potentially useful prognostic information, though this aligns with screening objectives.
3. **Generalizability**: Our dataset combines four specific institutional sources (Cleveland, Hungarian, Switzerland, Long Beach VA). While this improves diversity over a single source, the findings may not fully generalize to non-Western populations or younger cohorts without external validation.

C. Clinical Implications:

Gender-specific models should be considered for cardiovascular screening systems, particularly when sufficient gender-stratified training data is available. For females, simpler models with strong regularization may offer better generalization than complex ensemble methods. Clinicians should recognize that predictive feature importance differs by gender, suggesting potential value in gender-tailored risk assessment protocols.

D. Future Work:

1. Expand female sample collection to achieve parity with male sample size
2. Investigate severity prediction (multi-class classification) for prognostic applications
3. Explore deep learning approaches with larger datasets

V. Contribution

As the sole contributor to this project, the author was responsible for all aspects, including:

- Literature review and problem formulation
- Dataset acquisition, cleaning, and preprocessing
- Experimental design and implementation
- Model development, training, and optimization
- Results analysis and interpretation
- Report writing and documentation

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STATEMENT ON THE USAGE OF AI

Generative AI tools were utilized to refine specific code segments and enhance the academic tone of the report. The core conceptualization, experimental design, and initial code implementation were performed exclusively by the author. All results and conclusions were verified by the author.